



Curve Master Docs



Tips: Join our Discord for support and showcasing your work!



Overview

Curve Master is a powerful workflow tool for animation, with functions such as bulk curve editing and keyframe manipulation. It bridges the gap between motion design software and Unity's animation tools.

Suitable for various project including UI animations, in-game cutscenes, cinematic rendering and more.



Compatibility

Curve Master works with Unity 2018.4 or higher.

Tested Unity Version:

- 2018.4
- 2019.4
- 2020.3
- 2021.3
- 2022.3
- 2023.2
- 6000.0



Links

| [Documentation](#) | [Tutorials](#) | [Discord & Support](#) |

Purchase in Asset Store

<https://assetstore.unity.com/packages/slug/252505?aid=1100lvWv6>

Purchase in Gumroad

<https://store.yan-k.tv/l/CurveMaster>

Purchase in Booth

<https://yan-k.booth.pm/items/4872279>

Features

- User-friendly GUI
- Bulk edit animation easing curves quickly and easily.
- Read and apply selected motion curves to other keyframes.
- Apply curve for ease in, ease out, or both in&out tangent
- Manipulate keyframes, such as time offset, alignment, and value reversal.
- Support Curves in Particle System and C# AnimationCurve (New in v1.2.0)
- Support Animation, Animator and Timeline keyframes.
- Works in both Dopesheet mode and Curves mode
- Create and share your own animation curve library.
- Full C# source code

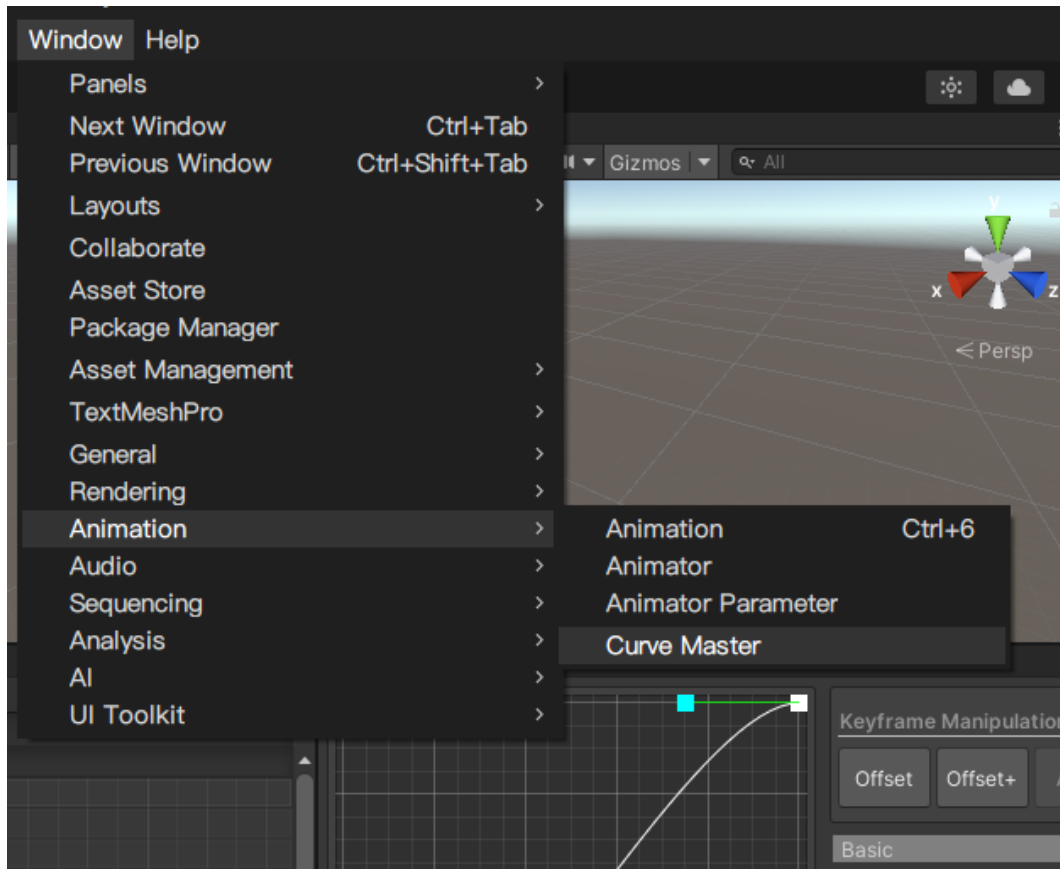
Battle Tested

How it started?

As a motion designer with over 10 years of experience, the development of this tool is to make animating in Unity faster and bridge the gap between motion design software (such as After Effects) and Unity's animation tools.

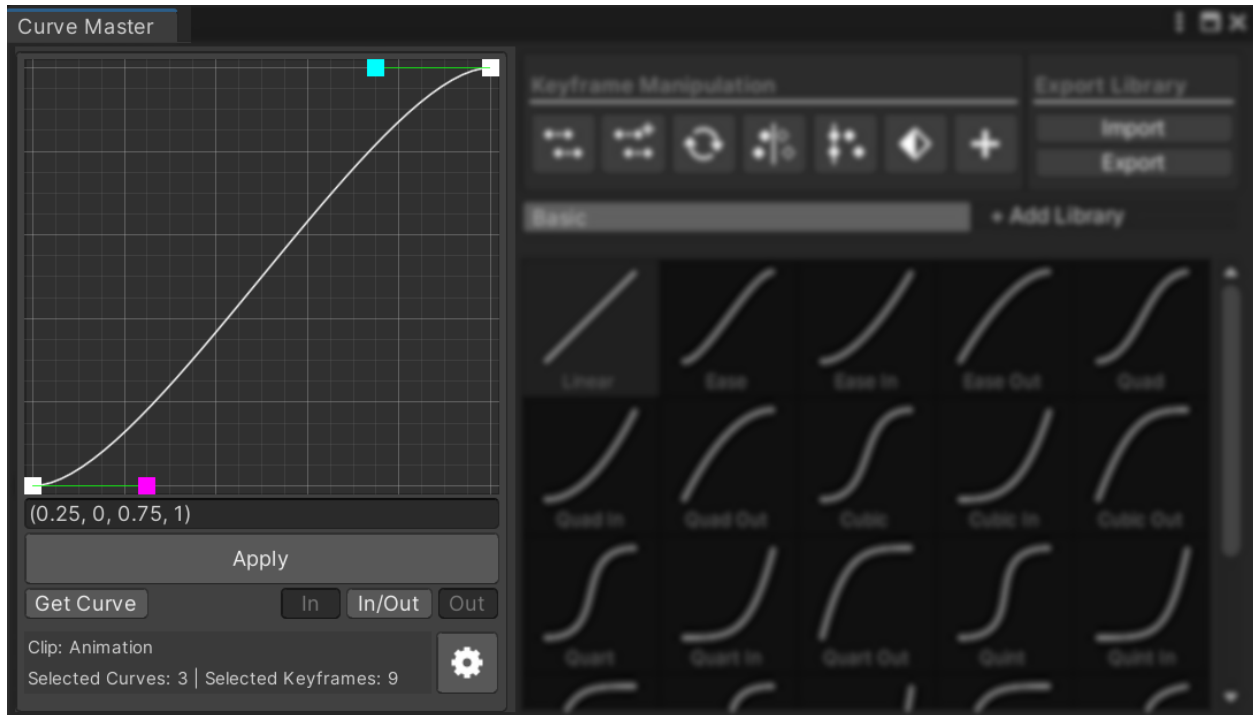
Curve Master has been battle tested in various project genres, including UI animations, in-game cutscenes and cinematic rendering to ensure it works well in different usage scenarios.

Usage



Open Curve Master GUI from Window → Animation → Curve Master

Curve Window



- Main Curve Area

Adjust curves using the GUI by dragging the handles, or manually enter curve values in the text box at the bottom. Click the Apply button to apply the curve to all selected curves.

- Get Curve Button | [Example Video](#)

Load the selected curve into the curve adjustment window. You can then apply the same curve to other keyframes or save it as a preset in the library.

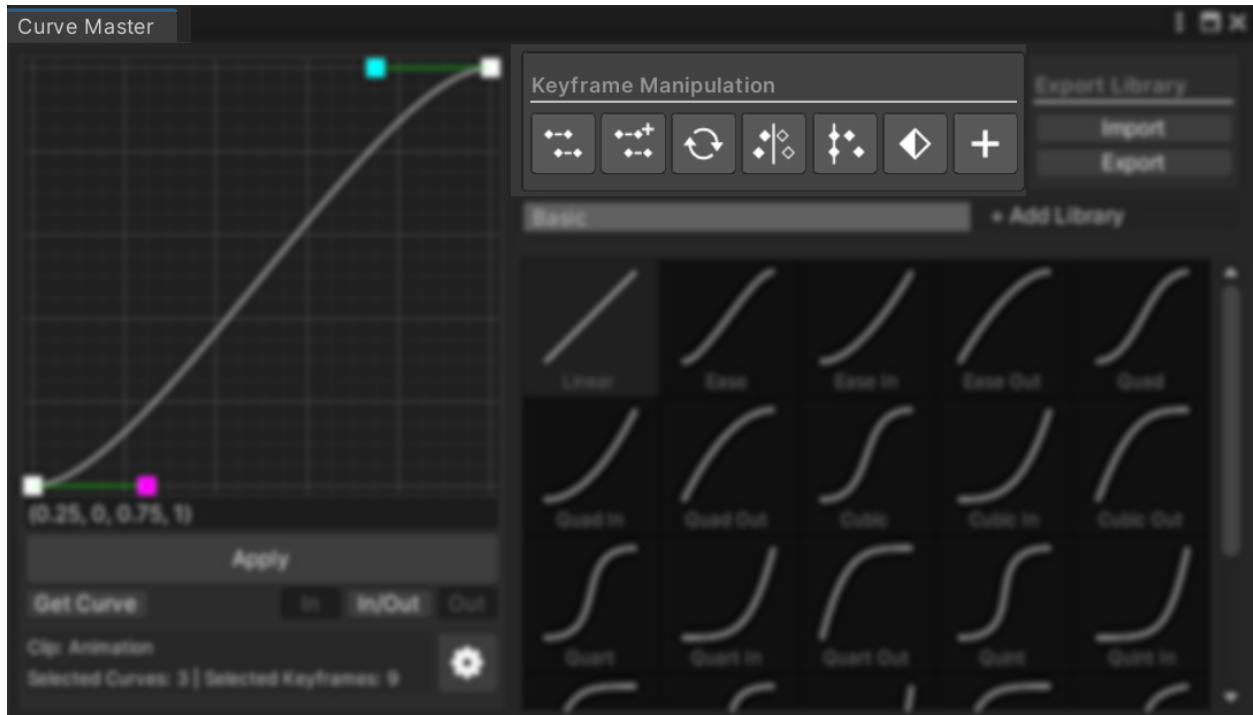
- In/Out Toggle

Apply the curve to the left handle (in tangent), the right handle (out tangent), or both.

- Info

Displays the current focused animation clip window, selected curves, and selected keyframes.

Keyframe Manipulation



- Offset | [Example Video](#)

Offset the selected keyframes by 1 frame across different gameobjects. The offset will follow the order of your keyframe selection.

- Offset+ | [Example Video](#)

Offset the selected keyframes by 1 frame for sub-properties, such as the XYZ values of a Vector3 or blendshapes within the same gameobject. The offset will follow the order of your keyframe selection.

- Reverse | [Example Video](#)

Reverse the value and tangent of selected keyframes.

- Mirror | [Example Video](#)

Duplicate the selected keyframes with reversed values.

- Align | [Example Video](#)

Move the selected keyframes to the current time. If multiple keyframes are selected, it will align the first keyframe you selected and maintain the timing gap for the other keyframes.

- Set Key | [Example Video](#)

Set the keyframe values of multiple properties to a custom value.

- Add Property | [Example Video](#)

Add properties from the selected object to the current animation clip with existing values.

(Currently supported: Position, Rotation, Scale, Transform, Blendshape)

Library



- Library Import/Export

You can import or export the currently selected library in .json format. Each library contains a GUID as a unique identifier. This means if you update your library and export it to share with others, they can "replace" their existing library with the same GUID.

- Library Tabs

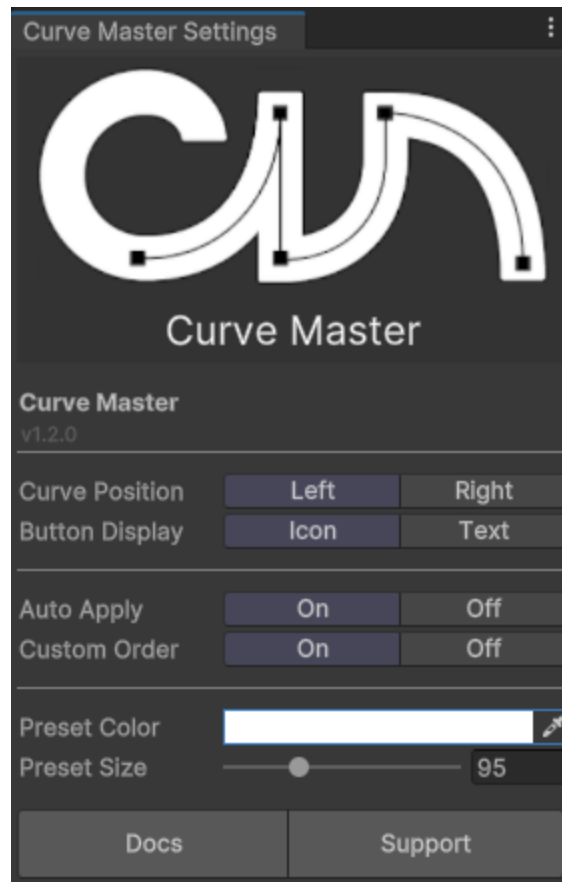
Click the + Add Library button to create a new library, or right-click on a library tab to rename or remove a library.

- Curve Presets

Click on any curve preset to load it into the main curve window on the left. The

large + button saves the current curve as a preset. You can right-click on a preset to overwrite, rename, or remove it.

Settings



- Curve Position

Change the curve window to be on the left side or the right side.

- Auto Apply | [Example Video](#)

Automatically apply curves to selected keyframes.

- Button Display

Switch between icon buttons or text buttons in "Keyframe Manipulation" section.

- Custom Order | [Example Video](#)

(Note: This may reduce performance when selecting a large number of keyframes)

Switch between top-to-bottom order or custom selection keyframe order for Offset and Offset+.

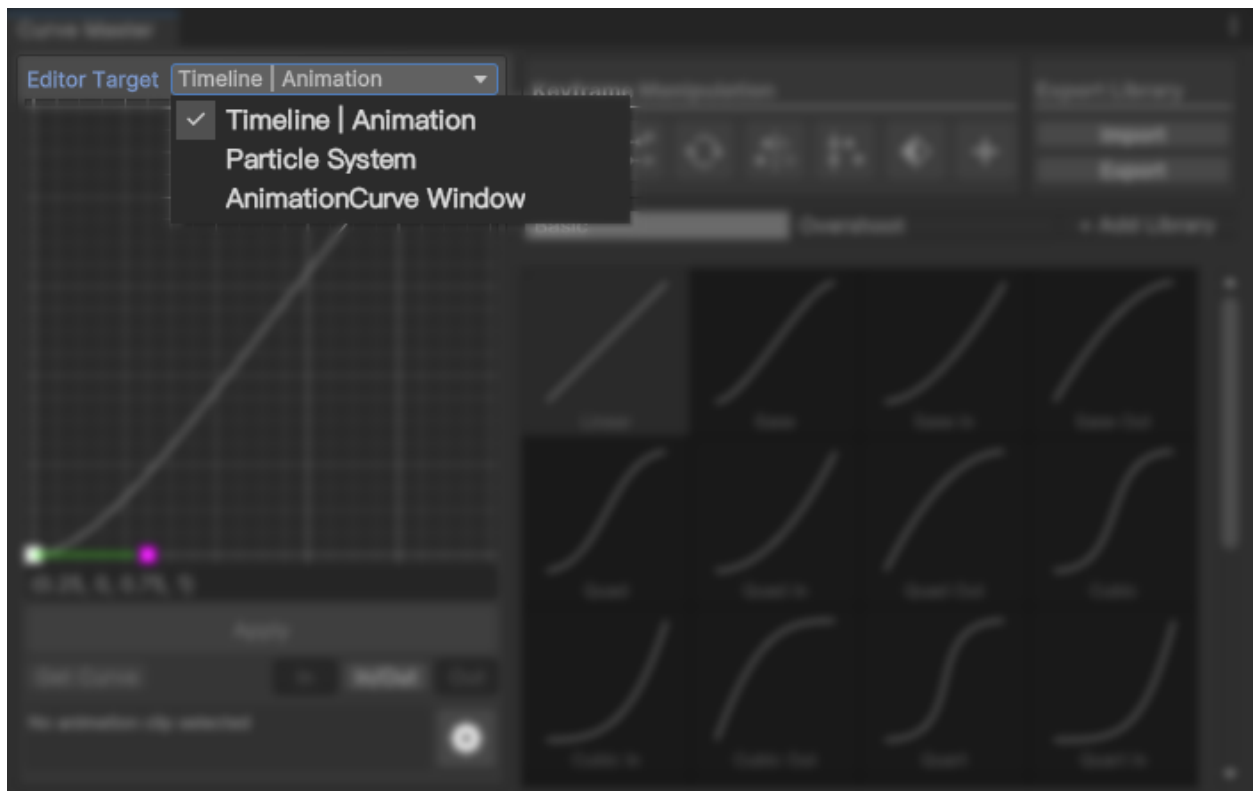
- Preset Color

Change the curve colors in the preset window.

- Preset Size

Change the curve size in the preset window.

Editor Target (New in v1.2.0)



- Timeline | Animation

This mode reads the selected curves and keyframes from the "Animation" window. It supports anything that reads AnimationClip, such as Timeline, Animator, and Animation.

- Particle System

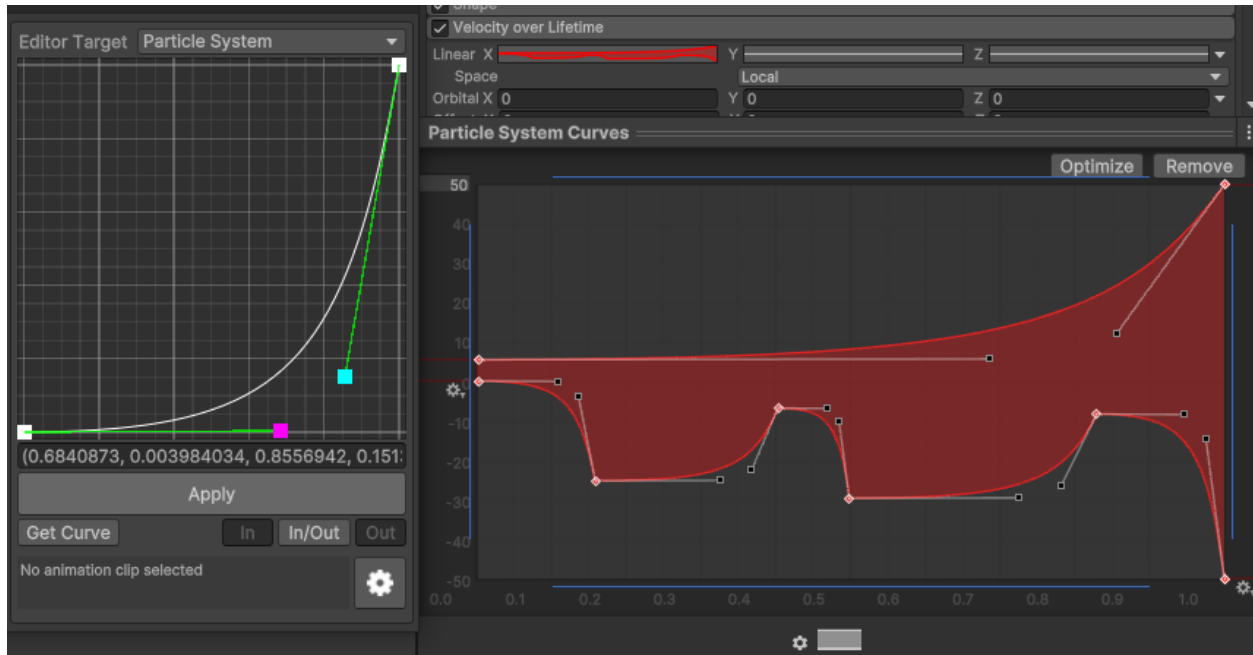
This mode reads the selected curve from the "Particle System" component. Use this mode to edit curves in the Particle System.

- AnimationCurve Window

This mode reads the selected curve in the AnimationCurve pop-up window. Use this mode to edit curves in your AnimationCurve property.

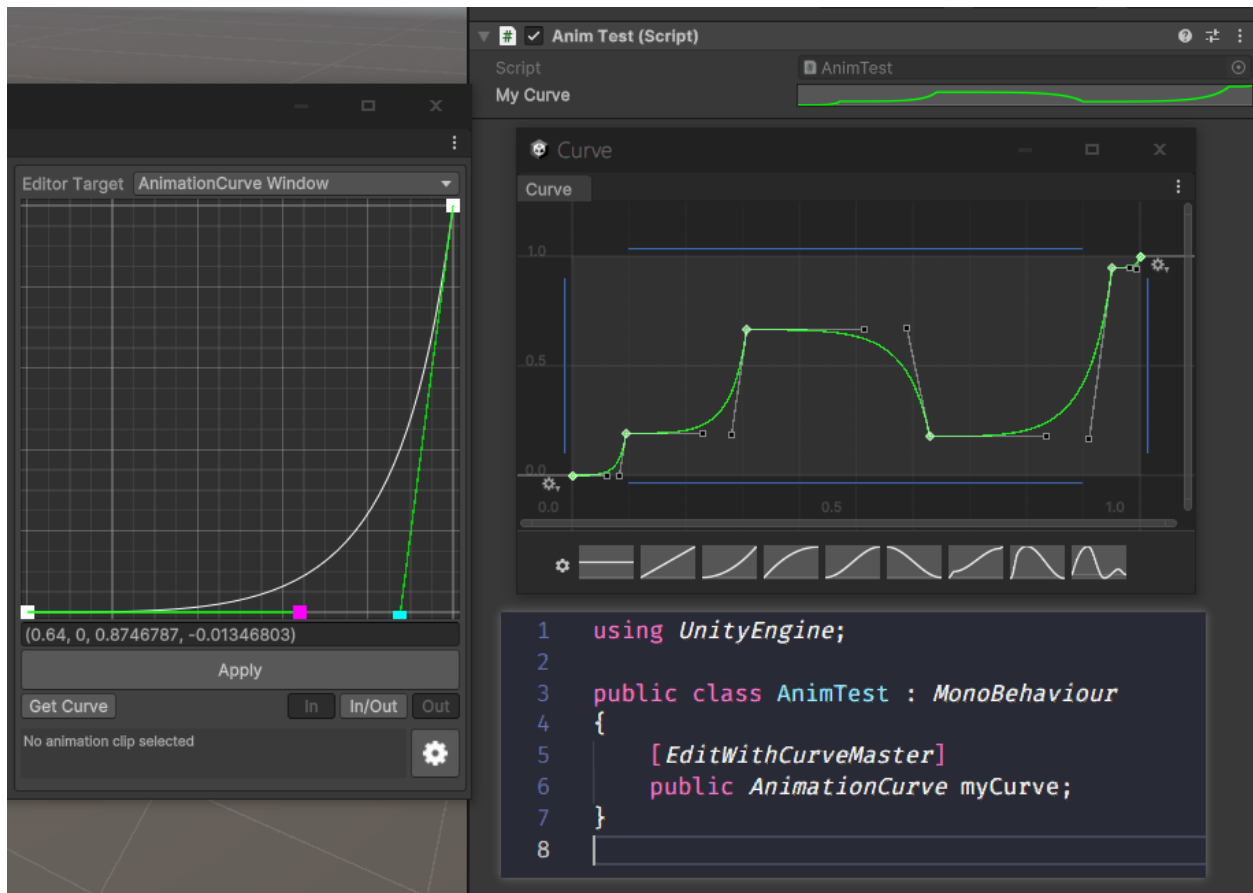
(▲ Only works with AnimationCurve properties that have the [EditWithCurveMaster] attribute)

Particle System Support (New in v1.2.0)



- When selecting "Particle System" in Editor Target, you will be able to get curves, apply curves, and save curves to the Particle System.

C# AnimationCurve Support (New in v1.2.0)



- Add the [EditWithCurveMaster] attribute to the AnimationCurve property you want to edit.

- When selecting "AnimationCurve Window" in Editor Target, you will be able to get curves, apply curves, and save curves to the AnimationCurve window.

▲ Normally, the AnimationCurve window will disappear when you click outside of the pop-up window. However, AnimationCurve properties with the [EditWithCurveMaster] attribute open a new pop-up window that stays on screen, allowing you to easily modify curves.

▲ FAQ & Troubleshooting

- Selected curves got reset and deselected.

Turn off the red recording toggle in Animation and Timeline window before doing curve adjustment.

- **"Get Curve" shows weird value.**

You might have selected too many keyframes at once, try only select 2 continuous keyframes at once.

- **Curve didn't apply to Particle System**

You need to select Particle System in "Editor Target" first so it can read selected curves. If you already selected curves in different Editor Target mode, you will need to de-select and re-select curves in order for the tool to read them.

- **"AnimationCurve" window disappear when click outside of pop-up window**

Make sure AnimationCurve have [*EditWithCurveMaster*] attached to it, so it won't be close automatically.



Credits

Yan-K (Yan-K Media Works)

[@YanKMW](#)

Dilaura (Calcatz Games)

[@dilaura_exp](#)

Itoshiki

[@itoshiki_s](#)



Special Thanks and Inspirations

cubic-bezier.com by Lea Verou

<https://cubic-bezier.com/>

Flow for After Effects by Rendertom

<https://aescrpts.com/flow/>

Motion Manager for Cinema 4D by Kengo Ito

<https://aescrpts.com/motion-manager/>

UniEaseCopy by Shuto

<https://github.com/shutosg/UniEaseCopy>



Changelog

1.2.0 (1-Year Anniversary Update, Coming Soon!)

⚠ Attention:

- This version contains breaking changes - Please REMOVE whole folder of old version before update to 1.2.0.
- Don't forget to BACKUP your preset library before deleting old version!

Fix:

- Reverse function now also reverse tangent of the curve.
- Preset color now unified in different Unity versions.
- "Custom Selection Order" in settings menu now has same style as other settings.
- Update Curve Master won't overwrite your preset anymore.

New:

- Custom preset color
- Custom preset size
- Support apply curves to "Particle System"
 - Added "Particle System" as editor target on top of the curve
- Support apply curves to "AnimationCurve"

- Added "Curve Window" as editor target on top of the curve

1.1.2

Fix:

- NullReferenceException error when animation window is not open.

1.1.1

New:

- Random value mode in "Set Key", now featured "Fixed Value", "Absolute Random" and "Relative Random".

Fix:

- Rotation in "Add Property" not animatable because it uses quaternion instead of euler angles

1.1.0

Beta ended, first public version.

New:

- Settings UI.
- Curve window position switch.
- Auto Apply.
- Keyframe Manipulation - Icon buttons.
- Keyframe Manipulation - Set Key.
- Keyframe Manipulation - Add Property.

Fix:

- Some keyframe won't get offset when keyframe gap is too close

Changes:

- Optimization, editor performance improved drastically.

1.0.6

Fix:

- Entire UI got disabled when no keyframe selected.
- Changed a function that only works in 2020+

1.0.5

Fix:

- Imported library didn't get saved without change.
- Mirror paste error when paste to existing keyframe.
- Reverse value error when selection contain objects with only 1 keyframe.

1.0.4

Changes:

- Added documentations
- Added tutorials
- Code clean up

1.0.3

Fix:

- Library name UI overflow
- Can't manually type negative value
- Apply curve to keyframes with same value

Changes:

- Added remove confirmation dialog
- Added Offset+ for property on same object

1.0.2

Fix:

- Curve UI disappear when read curve from 2 keyframes with same value

- Curve info doesn't update when click on a preset

Changes:

- Compatible with 2018.4+

1.0.1

Fix:

- Keyframe info doesn't show in dopesheet mode
- Reverse makes static value NAN
- Mirror reset curves in selected property

Changes:

- Mirror and Reverse work in dopesheet mode
- Automatically uses weighted tangent
- Make Align snap first selected keyframe
- Make Offset not affect sub-property

1.0.0

internal version with basic functionality.

Get Curve

Offset

Offset+

Reverse

Mirror

Align

Set Key

Add Property

Auto Apply

Custom Selection Order